

Metabarcoding Analysis Workflow

- Upstream Analysis

Miguel Semedo (msemedo@ciimar.up.pt)

2026/05/05

Metagenomic and bioinformatic tools for deep-ocean observation and diversity studies

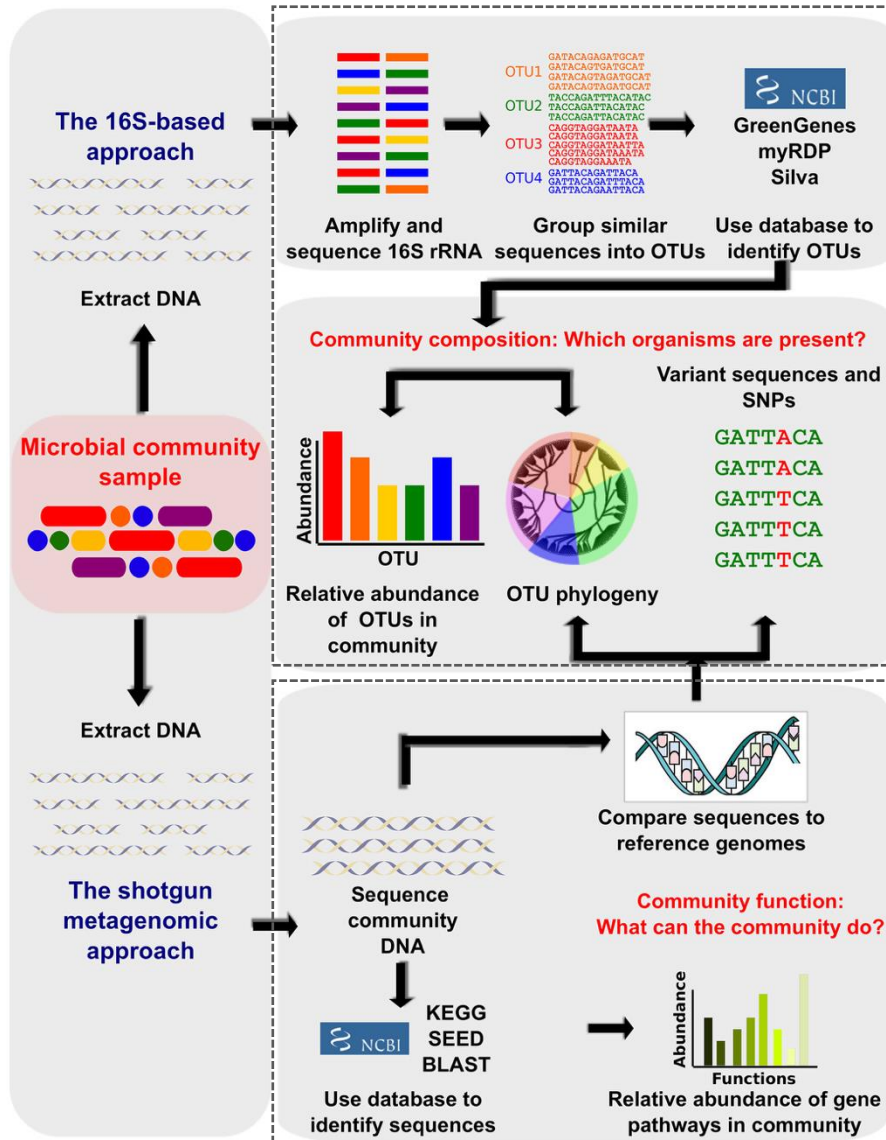


OUTLINE

1. Metabarcoding and metagenomic lecture catch-up and questions
2. Metabarcoding (focus on 16S rRNA gene sequencing)
3. Workflow overview (focus on upstream analysis)
4. Practical Example of the Upstream Processing (maybe for hands-on session)

1. Metabarcoding and Metagenomics Lecture Catch-up

Targeted and Untargeted Metagenomics



Targeted Metagenomics / Metabarcoding / Amplicon Sequencing

Marker genes amplification (e.g. 16S rRNA, others)

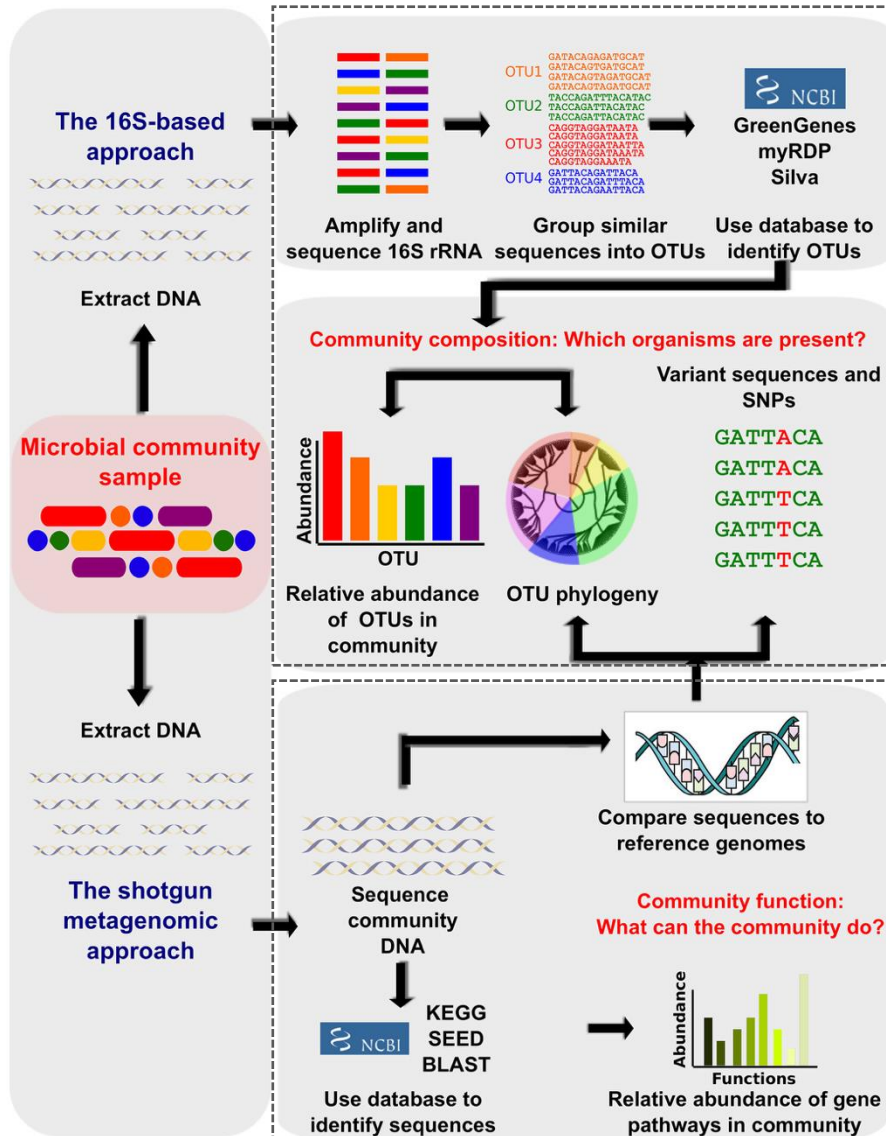
- Who's there (in general)?
 - With taxonomic markers: What can they possibly do (functional inference)?
 - With functional genes: Who's doing this specific function (potentially)?

Untargeted Metagenomics / Shotgun Sequencing / Whole Genome Sequencing

All **genomic** fragments (no PCR amplification)

- Who's there (in general)?
- What can they do (potentially)? Across infinite functions. **FUNCTIONS UNCHAINED!**

Targeted and Untargeted Metagenomics



Targeted Metagenomics / Metabarcoding / Amplicon Sequencing

MAIN LIMITATIONS:

- PCR-biased.
- Functional discovery limited to taxonomic classification (for 16S).

Untargeted Metagenomics / Shotgun Sequencing / Whole Genome Sequencing

MAIN LIMITATIONS:

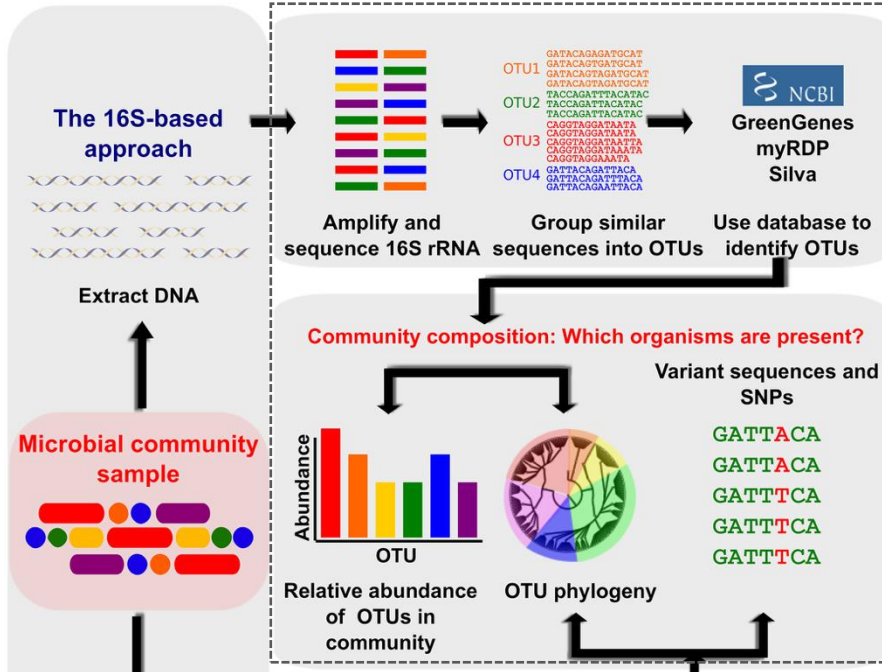
- Sequencing depth (maybe some rare species don't show up).
- Cost (€).

Also: Complementary / Different levels.
You can go for both applications

2. Metabarcoding/amplicon sequencing

Amplicon sequencing – What for?

Morgan and Huttenhower, 2012. PLoS Comput Biol 8(12):e1002808.



Targeted Metagenomics / Metabarcoding / Amplicon Sequencing

Marker genes amplification (e.g. 16S rRNA, others)

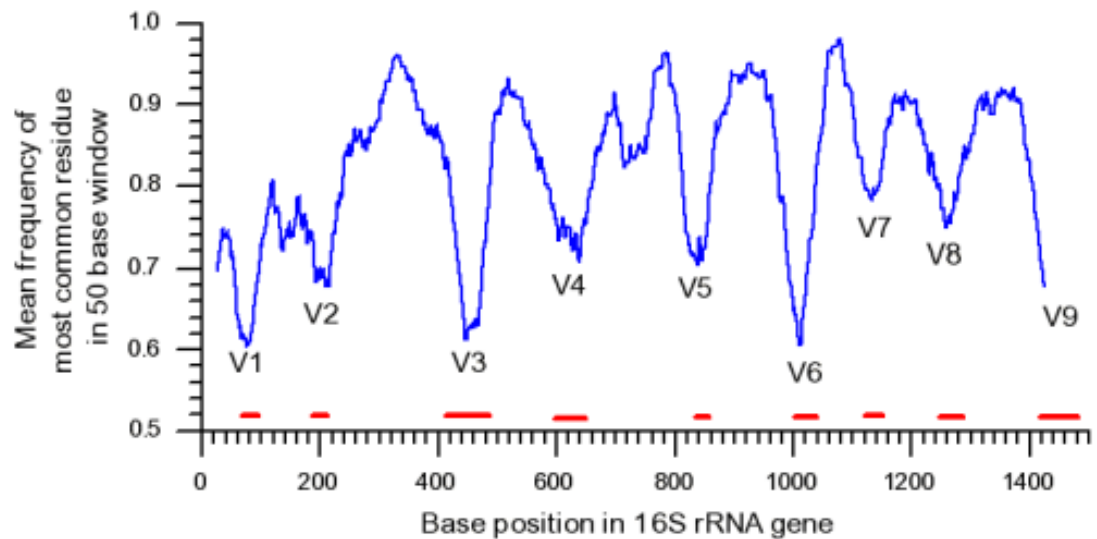
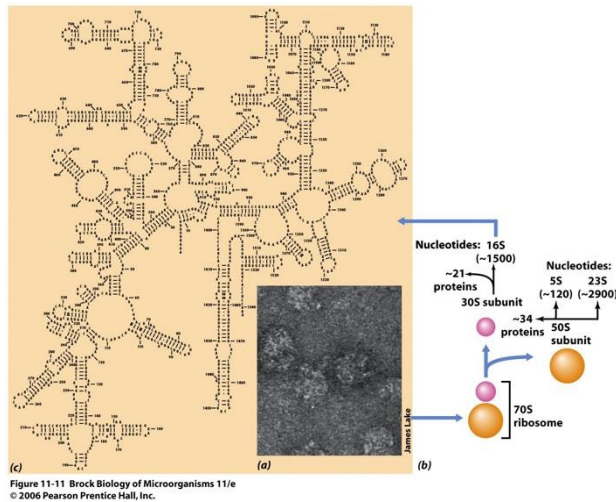
• WHO'S THERE (IN GENERAL)?

- With taxonomic markers: **What can they possibly do (functional inference)?**
- With functional genes: **Who's doing this specific function (potentially)?**

16S rRNA gene sequencing

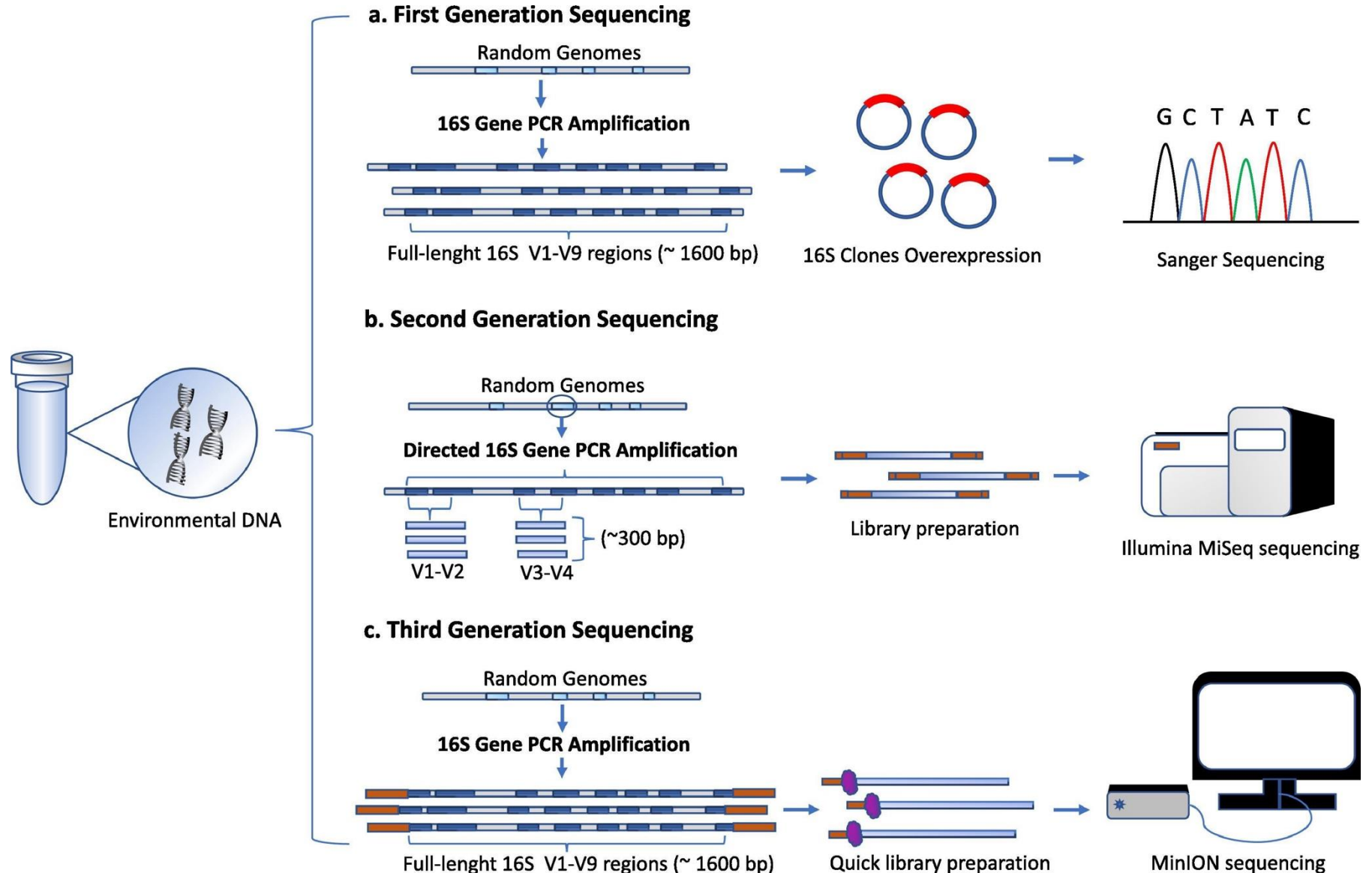
- **Taxonomy marker for prokaryotes (others for eukaryotes - can you name a few examples?).**
- How does composition and diversity vary across samples/conditions?
- What is the core microbiome of this particular sample type?
- What microorganisms are present exclusively in certain samples?
- What are the main biotic and abiotic interactions (can be specific – hypothesis driven)?
- What is the effect of a pollutant on bacterial diversity?

Why the 16S rRNA gene?

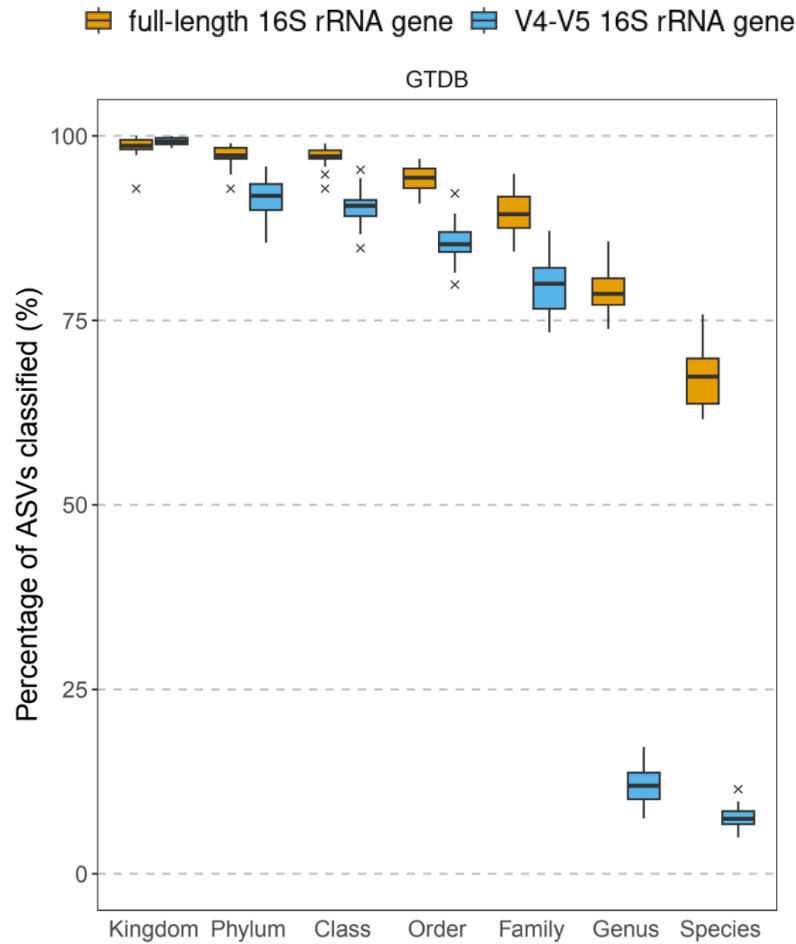


- RNA component of the small ribosomal unit in prokaryotes
- Universal marker for bacterial taxonomy
- Found in all bacteria and archaea (not single copy-gene in all cases – careful!). It can also be detected in eukaryotes. **This advantage can also be a problem, which one?**
- Different regions with different variability levels (highly conserved across all taxa and highly variable)
- Phylogenies derived from 16S rRNA gene tend to agree well with other conserved genes (evolution)

16S rRNA gene sequencing – “3 technology generations”



16S rRNA gene sequencing – short vs. long-read



Long-read sequencing (PacBio or ONT) tend to improve taxonomic assignment, especially at the lowest ranks (Genus, Species). Why do we still use short-read sequencing (Illumina)?

- High yield (number of reads per run)
- Low cost (vs PacBio, this may change o/t)
- Low error rate (vs ONT)
- Comparability with past studies and large datasets
- Difficulty to have long DNA fragments (sample quality)

Ultimately, it depends on your research needs...and sample type + opportunities/access to sequencing facilities.

16S rRNA gene sequencing – What degree of classification?

WHO'S THERE?

"The bacterial species problem"

Impossibility of using the traditional biological species concept (no sexual reproduction).

Prokaryotic "species" has been defined as:

- Strains with $\geq 70\%$ **DNA-DNA hybridization** and a difference in melting temperature of $< 5^\circ\text{C}$ (Wayne et al., 1987).
- Same species if their **16S rRNA genes are $\geq 97\%$ identical** (Stackebrandt and Goebel, 1994). Later reevaluated at 98.7%.
 - Observation: Different species with 99% similarity in the 16S rRNA gene may have very different physiologies (e.g. *Bacillus* species).

Still a "definition in progress"...

Circumventing the problem

Other terms: ecotype, phylotype, ribotype, oligotype, clade (different meaning), etc.

Operational Taxonomic Unit (OTU): usually defined as a cluster of organisms with **16S rRNA genes that are $\geq 97\%$ similar** (threshold may vary). Two main methods: closed-reference and *de novo* clustering.

Limitations of the OTU-based approach (clustering):

- Arbitrary dissimilarity threshold.
- No consistency converting to taxonomic levels.
- Distances within a taxonomic group may not be evenly distributed.
- Dataset-dependent (de novo method).
- Clustering is computationally intensive.

Circumventing the problem

More recently: **Amplicon Sequence Variants (ASV)**.

OPEN

The ISME Journal (2017) 11, 2639–2643
www.nature.com/ismej

PERSPECTIVE

Exact sequence variants should replace operational taxonomic units in marker-gene data analysis

Benjamin J Callahan¹, Paul J McMurdie² and Susan P Holmes³

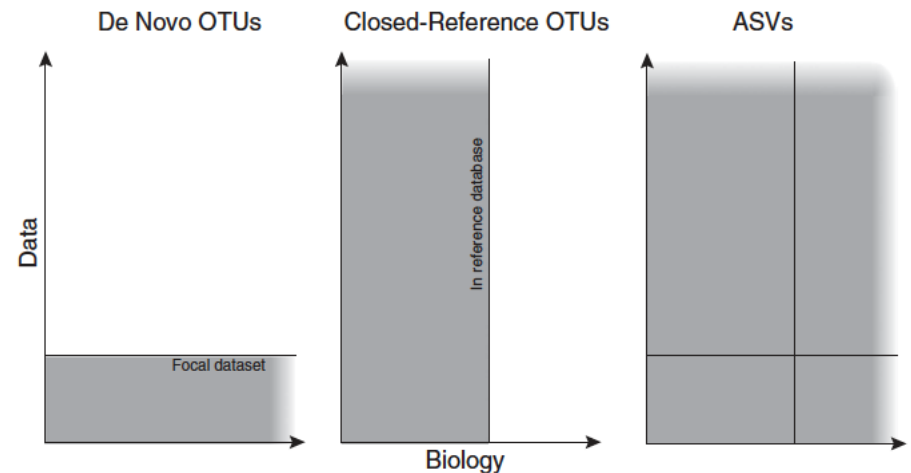
¹*Department of Population Health and Pathobiology, NC State University, Raleigh NC, USA;* ²*Whole Biome Inc, San Francisco CA, USA* and ³*Department of Statistics, Stanford University, Stanford CA, USA*

*ASV methods infer the biological sequences in the sample prior to the introduction of amplification and sequencing errors, and **distinguish sequence variants differing by as little as one nucleotide**. (In Callahan et al., 2017)*

Circumventing the problem

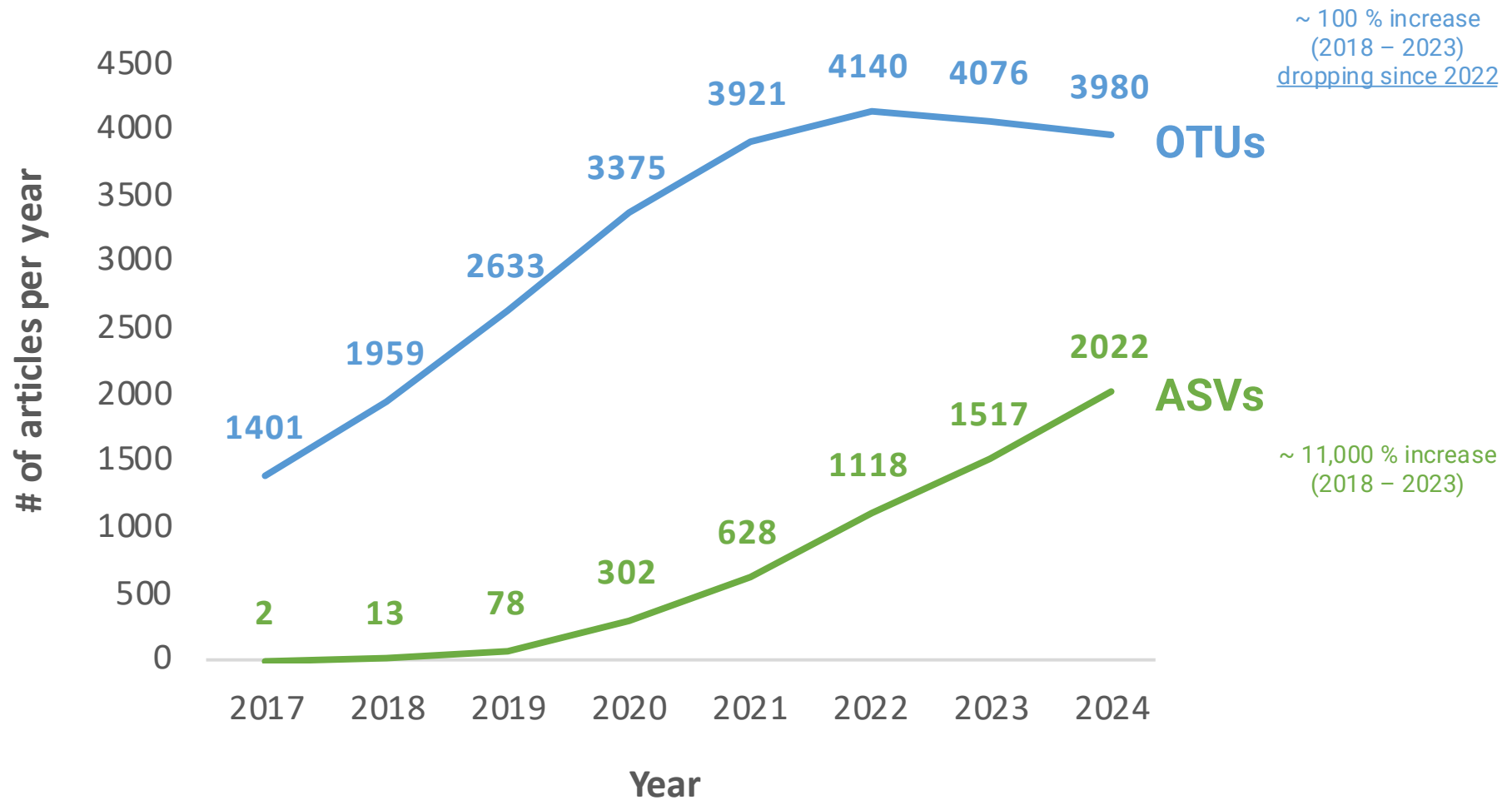
Main advantages of ASV vs. OTU:

- Higher resolution (obvious).
- Intrinsic biological meaning.
- Independent from reference database (compared to closed-reference OTU methods).
- Reusability (across different datasets – compared to *de novo* OTU methods).
- Reproducible in the future.



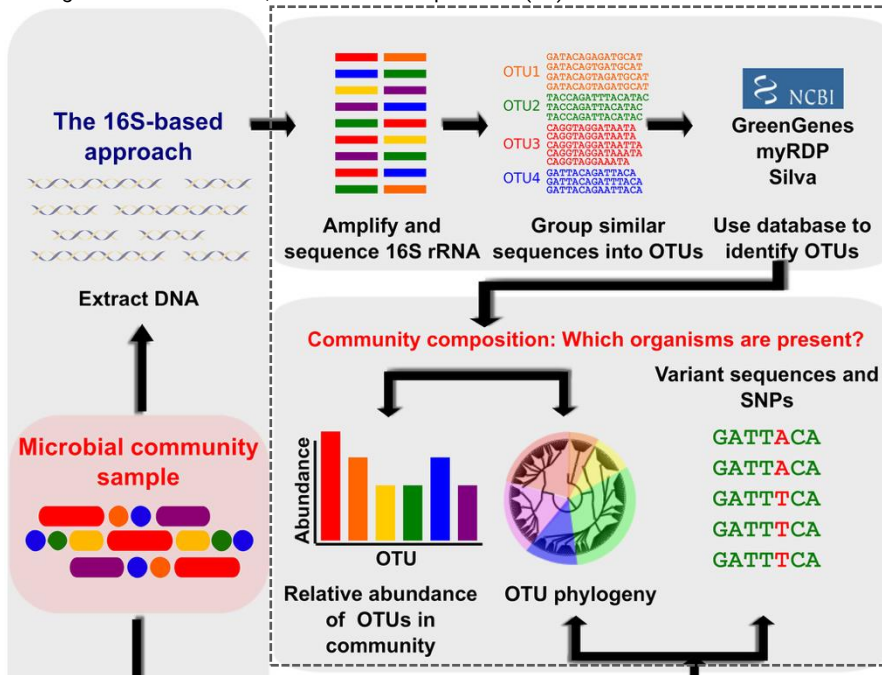
Callahan et al., 2017, The ISME Journal.

Current scientific use



Amplicon sequencing – What for?

Morgan and Huttenhower, 2012. PLoS Comput Biol 8(12):e1002808.



Targeted Metagenomics / Metabarcoding / Amplicon Sequencing

Marker genes amplification (e.g. 16S rRNA, others)

• WHO'S THERE (IN GENERAL)?

- With taxonomic markers: **What can they possibly do (functional inference)?**
- With functional genes: **Who's doing this specific function (potentially)?**

16S rRNA gene sequencing

- **Taxonomy marker for prokaryotes (others for eukaryotes - can you name a few examples?).**
- How does composition and diversity vary across samples/conditions?
- What is the core microbiome of this particular sample type?
- What microorganisms are present exclusively in certain samples?
- What are the main biotic and abiotic interactions (can be specific – hypothesis driven)?
- What is the effect of a pollutant on bacterial diversity?

2. Metabarcoding Workflow Overview

16S rRNA gene sequencing – typical workflow

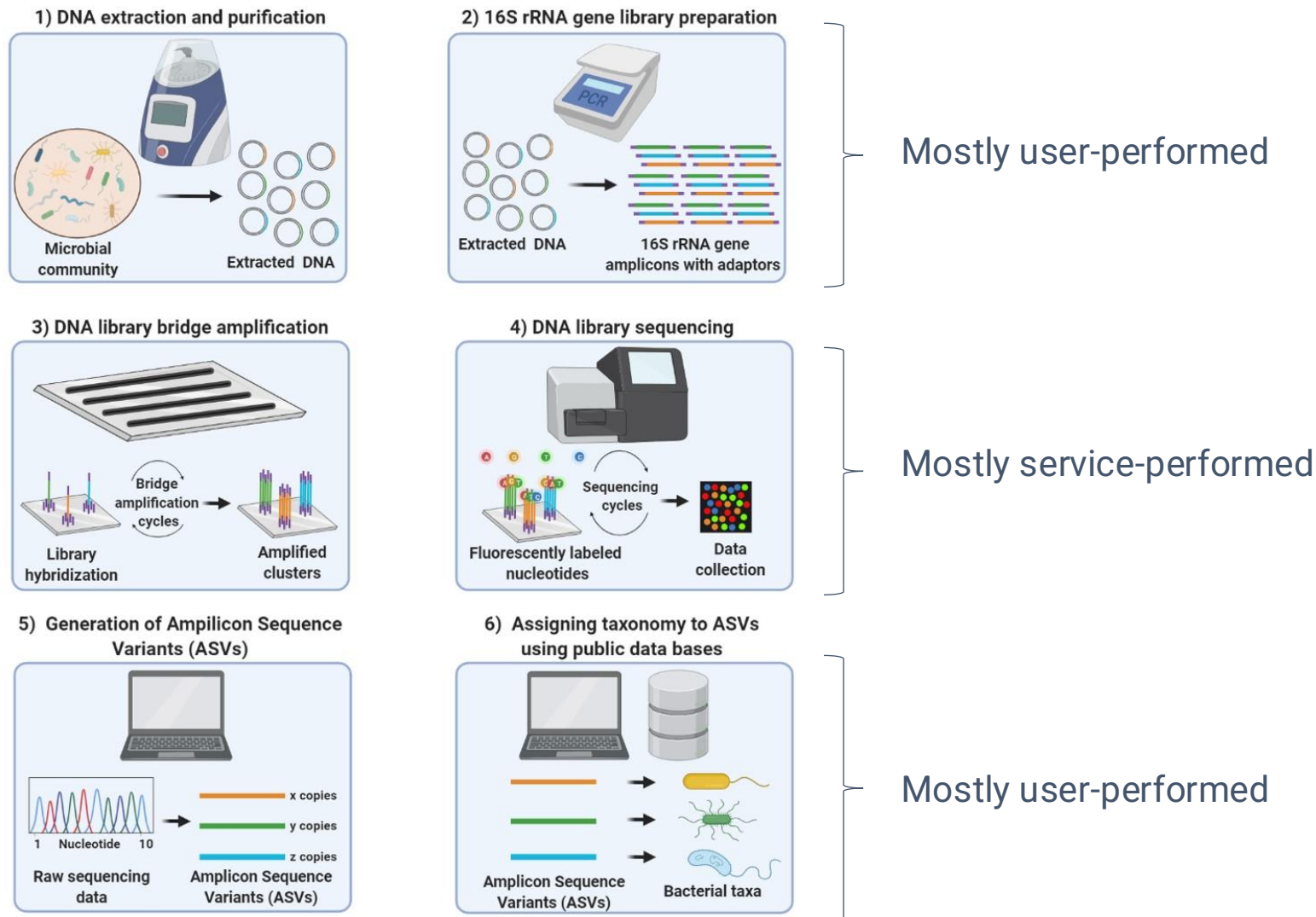


Diagram source: <https://www.ufz.de/index.php?en=48523>
(UFZ Helmholtz Center for Environmental Research)

Generated Data

Main output from any sequencing technology is a **FASTQ file**

- Text files that store biological information in the form of strings of nucleotides
- Formally defined in 2010
- Gold standard file format for storing and representing biological sequence data

@ and sequence identifier
raw sequence

— @M04577:12:000000000-D2N7B:1:1101:15525:1827 1:N:0:23
TCTTTCGCTGGCGCGTAAACAGCAAAAGCAGGTACTGATTCAAATTGTTTGTCTTTGTTCTTCTTTGCTTCATTCCTTTTTTCTTCTTTC
TTTCTTTTTTTTCTTTTTTTTTTTTTTTTTTTTCCCTTATTCTTTTCTTTTCTCTTTTTTTTCTCTCTTTTTTTTTTCTTTTTCTT
TTTTCTTTTTTCCTTTTCTTCTTTTTTTTTTTTTTTTCTTTTTTTTTTTTTTCTTTTTTTCCTTTT

metadata
Quality score per base call

— +
1AAAAFAAAADAGGGGGGEBBC00100010/B1AB22DD21222B1220002DDB1B0121221A11DD112D2222B11>01@22112BB1221111///0
11B111111//>/>/-----:/000000;000000;000000;9/00;0000:-9-///;///;9/----9/://9B//:/:/9///:9@///:;B//;///;/------9---9///9//;9-9----;/:/-9///:/

What is a quality core?

$Q_{\text{score}} = -10 \log_{10} p$

p = probability that some base was called incorrectly

Phred Quality Score	Probability of incorrect base call	Base call accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1000	99.9%
40	1 in 10,000	99.99%
50	1 in 100,000	99.999%
60	1 in 1,000,000	99.9999%

Table source: wikipedia.org

16S sequencing analysis workflow (from short reads)

Different softwares/pipelines/workflows have been used to process amplicon NGS data, such as Mothur, QIIME, and more recently DADA2 (R package), that applies the concept of “**amplicon sequence variants (ASVs)**”.

Package ‘dada2’

May 13, 2018

Type Package

Title Accurate, high-resolution sample inference from amplicon sequencing data

Description The dada2 package infers exact amplicon sequence variants (ASVs) from high-throughput amplicon sequencing data, replacing the coarser and less accurate OTU clustering approach. The dada2 pipeline takes as input demultiplexed fastq files, and outputs the sequence variants and their sample-wise abundances after removing substitution and chimera errors. Taxonomic classification is available via a native implementation of the RDP naive Bayesian classifier, and genus-species assignment by exact matching.

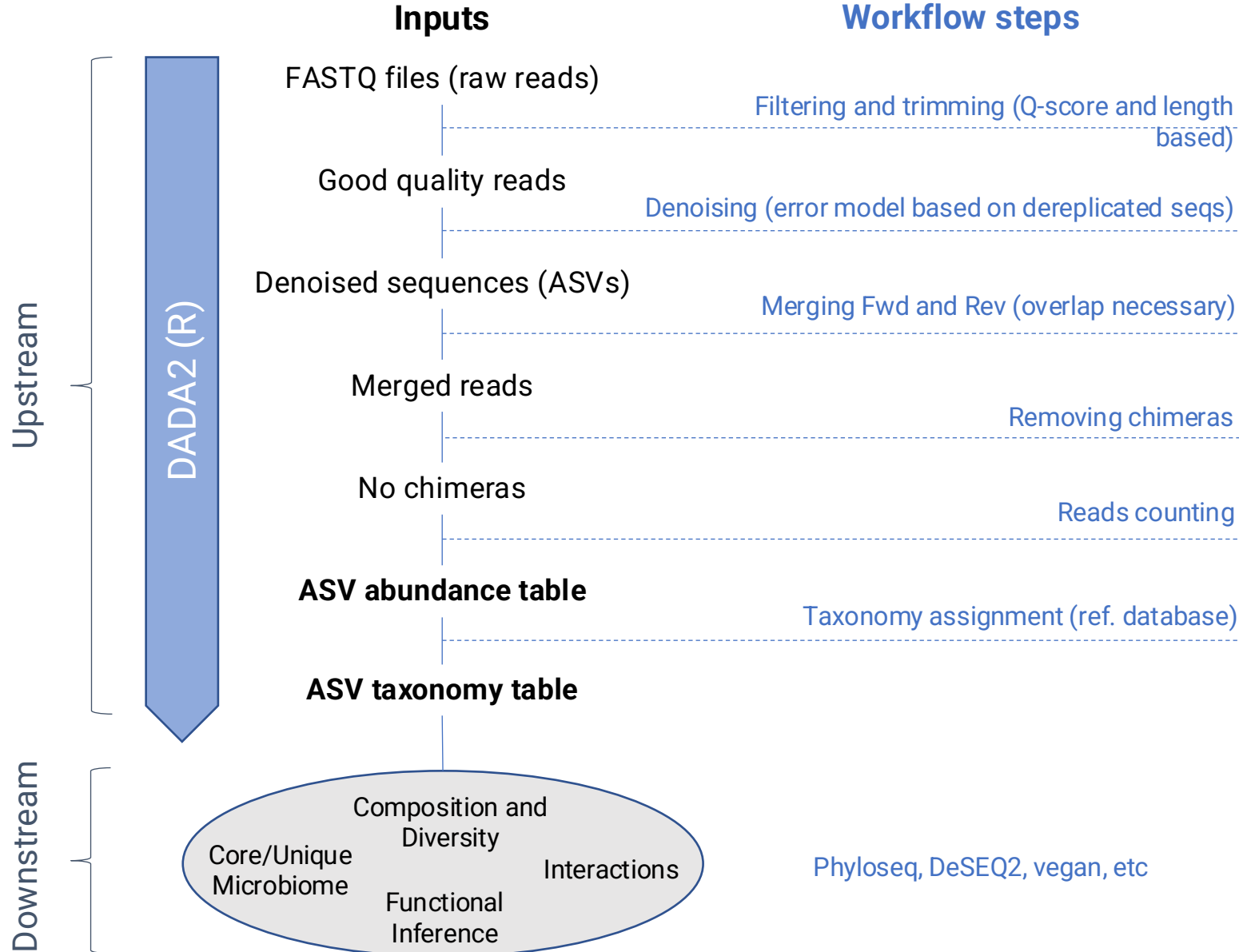
Version 1.9.0

Date 2018-1-30

Maintainer Benjamin Callahan <benjamin.j.callahan@gmail.com>

Author Benjamin Callahan <benjamin.j.callahan@gmail.com>, Paul McMurdie, Susan Holmes

16S sequencing analysis workflow (from short reads)



16S sequencing analysis workflow (from short reads)

WORKFLOW SUMMARY REPORT (UPSTREAM)

33 libraries	Input RAW sequences	Filtered + Denoised	Merged	non chimeras	Total Yield
Total	<u>5,734,065</u>	5,080,131	2,939,179	<u>1,972,593</u>	<u>34.4%</u>
Step Yield		88.6%	57.9%	67.1%	
Average/sample	173,760 ± 55,517	153,943 ± 49,754	89,066 ± 32,413	59,776 ± 20,666	

The balance between merging overlap and chimera removal is often sensitive (play with it!).

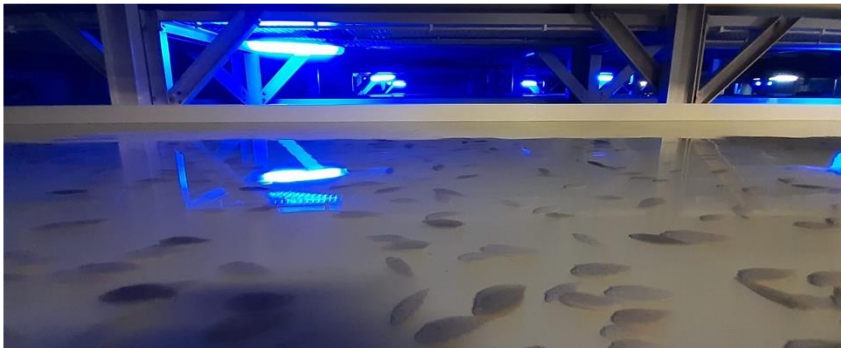
I like to work with a minimum of 30% total yield and 20,000 final number of high-quality sequences, but this is highly dependent on the study and research question. If some sample is under these criteria, I consider removing it from the downstream analysis.

3. Practical Example

16S sequencing analysis workflow (practical example)

Microbial Community Structure of Juvenile Sole (*Solea senegalensis*) Fed with Different Diets (manuscript in preparation)

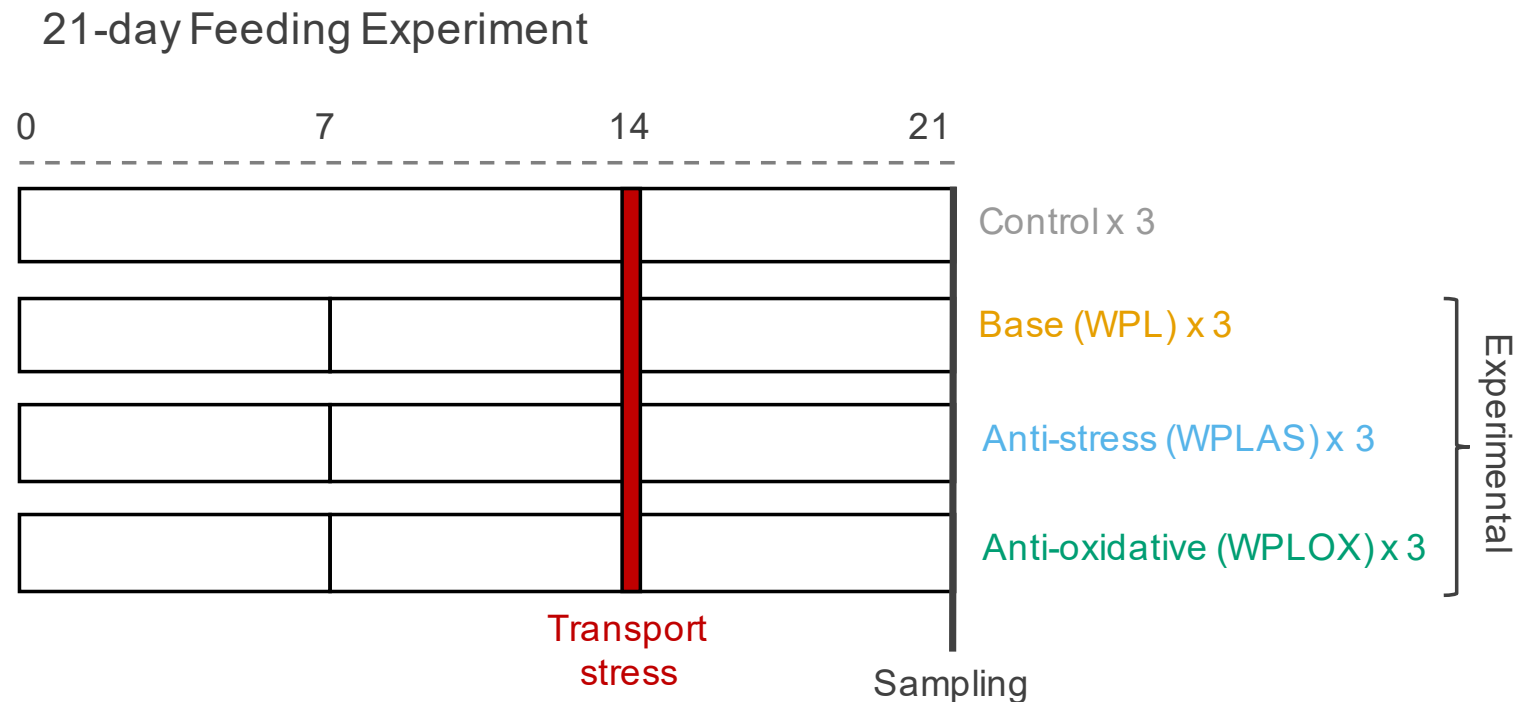
- 1) Investigate the microbial community structure across different fish body parts
- 2) Investigate the effects of diet composition in the microbial community structure of the different matrices



16S sequencing analysis workflow (practical example)

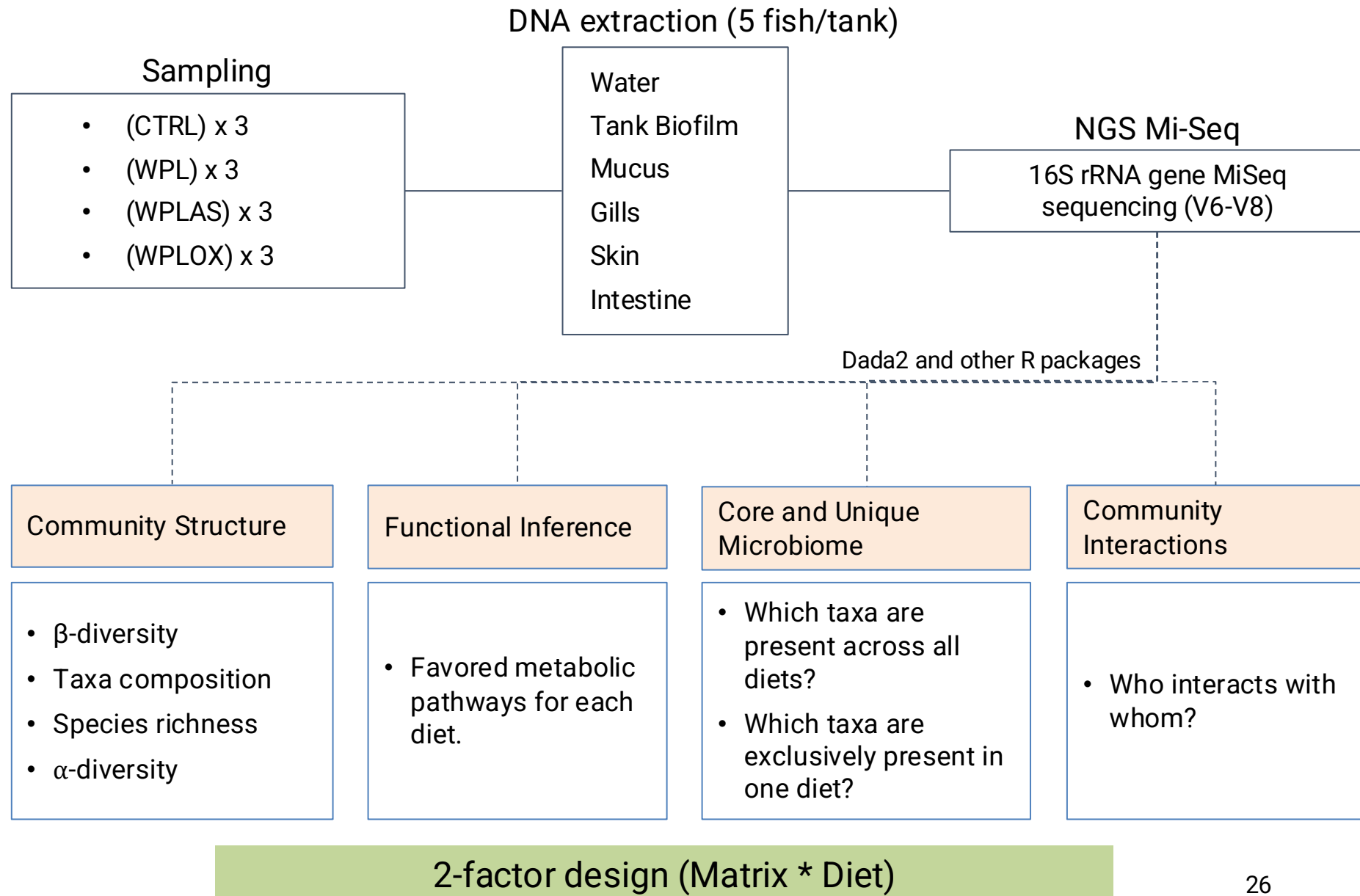
Microbial Community Structure of Juvenile Sole (*Solea senegalensis*) Fed with Different Diets (*manuscript in preparation*)

Experimental Design



12 tanks (4 diets x 3 replicates)

16S sequencing analysis workflow (practical example)



16S sequencing analysis workflow (practical example)

DADA2 (just the intestines for the workshop, n= 12)

Before start (checklist)

1. Check service provider report (raw, filtered, merged, etc.). Is the initial number of sequences good enough? Are adapters and primers removed?
2. Sequence files (chose which ones to use)
3. Metadata table (“map object”)
4. Get Log text file (**show**)
5. Your taxonomy reference database (Silva / GTDB / Greengenes)
6. Start a new R script (or open a trusted one)
7. Don't just trust (blindly) someone else's script
8. Start a new RStudio session (set working directory)

16S sequencing analysis workflow (practical example)

DADA2 run (just the intestines, n= 12)